

CY 547: Chemical Crystallography

Course Code	CY 547
Course Name	Chemical Crystallography
L-T-P-C	2-0-2-3
Prerequisites	Consent of Teacher
Intended for	M.Sc., Ph.D.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Unit 1** Crystallography overview, X-ray generation, properties of X-rays, crystal lattices, the unit cell and crystal systems, Bravais lattices, lattice planes and hkl indices, Braggs law, reciprocal lattice, structure factors and phase problem. [3Lectures]
- **Unit 2** Crystal symmetry, symmetry elements, space groups, asymmetric unit, Laue groups, systematic absences. [4Lectures]
- **Unit 3** Experimental methods, crystal growth, crystal mounting, four circle diffractometer, area detectors, data reduction, absorption correction. [4Lectures]
- **Unit 4** Structure solution and refinement, Patterson methods and Direct methods, least squares refinement, crystallographic R-values, refinement techniques. [6 Lectures]
- **Unit 5** Anomalous dispersion and Absolute Structure, chiral and polar space groups, disorders, twinning, space group errors. [5 Lectures]
- **Unit 6** Interpretation and presentation of results, crystallographic information file, checkcif, bond lengths and angles, torsion angles, Ortep plots, crystallographic literature and data bases. [6 Lectures]
- **Topics to be covered in practical sessions** [28 Lectures]
Using appropriate sample crystals, practical training will be given on the following topics: crystal selection and mounting, unit cell determination and setting up data collection strategy, data reduction and absorption correction using the software package provided, structure solution and refinement using SHELX/ provided package, preparation of cif file, checkcif and preparation of ortep diagram and other publication data of the structure. Brief training will be given in structure refinement of special cases like those with twinning, disorders etc. using standard structural data available on internet resources. Data base search.

Text Books

1. Werner Massa, **Crystal Structure Determination**, 4th Edition, Springer, 2010.

References

1. J. P. Glusker, M. Lewis M. Rossi, **Crystal Structure Analysis for Chemists and Biologist**, VCH, 1994.
2. M. M. Woolfson, **An Introduction to X-ray Crystallography**, 2nd Edition, Cambridge University Press, 1997.
3. M. F. C. Ladd, R. A. Palmer, **Structure Determination by X-ray Crystallography: Analysis by X-rays and Neutrons**, 5th Edition, Springer, 2012.
4. The SHELX-97 Software Manual.